

VIVEK SAI ALAJPUR

PRODUCTION ENGINEER | SMT | THT | NPI | MANUFACTURING SCALE-UP

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PROFESSIONAL SUMMARY

Production Engineer with hands-on experience in SMT and THT manufacturing, new product introduction (NPI), and high-volume production scale-up. Proven ability to reduce defects from 40–50% to below 10% and execute rapid production ramps (90,000 units in 40 days). Experienced in design for manufacturing (DFM), SMT programming, process stabilization, and production system development in startup environments.

PROFESSIONAL EXPERIENCE

Production Engineer - Trielectra LLP

May 2025 – Present

- Managed end-to-end production for multiple electronic products, ensuring high-volume output, quality, and process stability.
- Executed production of 50,000 Puppy House toy boards within 35 days, maintaining assembly consistency and throughput.
- Supported manufacturing of 15,000 drone receiver boards and 15,000 transmitter boards (externally designed) within 50 days, handling SMT/THT processes, debugging, and production stabilization.
- Led design, development, and mass production of audio-responsive control board (CACTUS toy product):
 - Scaled to 90,000 units dispatched in 40 days (ongoing), target 155,000 units
 - Reduced defects from 40–50% to 8–10% through DFM corrections and SMT process optimization
 - Resolved issues in audio signal amplification, noise, and circuit stability
- Developed SOPs and testing methodology to ensure repeatable production
- Drove prototype-to-pilot transition for drone receiver and transmitter boards (two models designed in-house), preparing designs for manufacturing readiness and pilot validation.
- Programmed and optimized SMT pick-and-place machines, improving placement accuracy and reducing assembly defects.
- Identified and corrected PCB issues including footprint errors, polarity mismatches, and pad geometry issues, preventing large-scale rework.
- Developed production SOPs, test processes, and workflow improvements, enabling consistent and scalable manufacturing.

Co-Founder & Hardware Engineer - Shoolin Automation Technologies

2023 – 2025

- Co-founded a smart home automation startup, developing hardware products from concept to functional prototypes.
- Designed single- and double-layer PCBs and fabricated prototypes using CNC-based PCB prototyping tools.
- Managed BOM creation, component sourcing, and cost optimization for small-batch production.
- Integrated ESP32, ESP8266, and Arduino-based systems with relays, sensors, and wireless communication.
- Performed hardware debugging, testing, and field validation to improve reliability.

KEY ACHIEVEMENTS

- Reduced production defects by >75%, significantly improving yield and reducing rework.
 - Executed high-volume production including 90,000 units in 40 days and multiple product lines simultaneously.
 - Established SOPs and testing frameworks from scratch in a startup environment.
 - Acted as a key interface between design, manufacturing, and assembly teams.
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CORE SKILLS

- SMT Assembly
 - THT Assembly
 - New Product Introduction (NPI)
 - Design for Manufacturing (DFM/DFA)
 - PCB Design Review & Gerber Validation
 - Pick-and-Place Programming
 - Yield Improvement & Process Optimization
 - Root Cause Analysis
 - Production Planning & Execution
 - SOP Development & Testing
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TECHNICAL SKILLS

Hardware Design & Validation:

- Analog Circuit Design, Audio Signal Amplification, Signal Conditioning, PCB Design & Layout Validation

Manufacturing & Hardware:

- SMT, THT, Soldering, Rework, Inspection, PCB Prototyping

Tools:

- EasyEDA, KiCad, DipTrace, FlatCAM, Mach3Mill

Platforms:

- ESP32, ESP8266, Arduino, STM32 (Basic)
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EDUCATION

Bachelor of Engineering (Electronics & Communication)

Balaji Institute of Technology & Science | 2021 – 2024

Diploma (Electronics & Communication)

Vemuganti Manohar Rao Polytechnic | 2018 – 2021